

Title

me

Max Planck Institute for Gravitational Physics

(Dated: September 21, 2026)

abstract

I. INTRODUCTION

Gravitational waves provide a unique observational window into astrophysical and cosmological phenomena. Since the first direct detection in 2015 [1], the LIGO-Virgo-KAGRA (LVK) collaboration has observed dozens of compact binary coalescences in the 10 Hz to 10 kHz frequency band [1]. Accessing gravitational waves at millihertz frequencies requires a space-based detector to overcome terrestrial seismic and Newtonian noise and to provide sufficiently long interferometric baselines. The Laser Interferometer Space Antenna (LISA), a joint ESA–NASA mission scheduled for launch in the mid-2030s, is designed to operate in the 0.1 mHz to 1 Hz band [2].

The LISA constellation consists of three spacecraft arranged in an approximately equilateral triangle with arm lengths of about 2.5 million km. Laser beams exchanged between the spacecraft form a constellation-scale interferometer. Unlike ground-based detectors with effectively equal arms, the LISA arm lengths vary by up to about 1.2% due to orbital motion, preventing direct cancellation of laser frequency noise. Although cavity pre-stabilisation reduces laser frequency noise substantially, it remains several orders of magnitude above the displacement noise budget required for gravitational-wave detection [1]. To suppress this noise, time delay interferometry (TDI) has been developed as a post-processing technique that synthesizes virtual equal-arm interferometers by appropriately time-shifting and combining phasemeter measurements.

In addition to laser frequency noise, LISA measurements are affected by clock noise originating from the independent ultra-stable oscillators (USOs) carried by each spacecraft. Since all phase measurements are referenced to these clocks, timing jitter couples directly into the measured heterodyne beatnotes. Space-qualified USOs achieve Allan deviations of order 10^{-13} at one-second averaging times [3], which is insufficient for the picometer-level test-mass position sensitivity required by LISA. To mitigate this effect, LISA employs additional clock-noise transfer measurements—implemented via modulation sidebands exchanged between spacecraft.

TDI has been extensively studied theoretically [4] and validated through numerical simulations [5]. Such simulations, however, generate the measurements from the same model that the analysis pipeline assumes, and therefore test the implementation of the model rather than the model itself. Testing the model requires data produced independently of it, by a physical realization of the mea-

surement chain. Experimental verification of this kind is still developing.

A major milestone was the University of Florida Laser Interferometry Simulator (UFLIS), which demonstrated that second-generation TDI combinations can suppress laser frequency noise by ten orders of magnitude [4]. The work presented here is a continuation of the UFLIS approach with the addition of independent clock sources for each spacecraft, largely enabled by developments in FPGA technology. Other experiments have also contributed important demonstrations relevant to TDI and clock-noise removal. The setup by de Vine et al. demonstrated laser frequency noise suppression at a level of roughly nine orders of magnitude and clock-phase noise suppression of about four orders, although without accounting for realistic photon light-travel times. The LISA on Table experiment in Paris more recently demonstrated laser-noise removal using direct digital synthesis (DDS)-generated signals and electronically imposed delays. The Hexagon experiment in Hannover has provided strong results for clock-noise removal and synchronization.

Here, we present an overview of an ongoing effort to develop a TDI testbed that combines realistic laser frequency noise, programmable delay lines, and independent clock sources within a single experimental framework. The testbed focuses on reproducing the dominant noise couplings of the LISA metrology system—specifically, laser frequency noise and clock noise—while preserving the structure imposed by unequal and time-dependent delays. Crucially, the same delay lines that impose the inter-spacecraft light-travel times also allow physically consistent gravitational-wave strain signals to be injected directly into the measurement chain. The testbed can therefore be used not only to demonstrate noise suppression, but to verify that a known gravitational-wave signal survives TDI processing of hardware-generated data with the expected amplitude and phase.

An initial purely electrical implementation centered around field programmable gate array (FPGA)-based digital delay lines has already been demonstrated in Ferguson et al. [REF]. The developments discussed here extend that architecture by incorporating an optical front-end with physically representative laser noise and independently running spacecraft clocks. This preserves the essential timing, noise, and signal-propagation structure of the LISA one-way measurements, allowing TDI combinations and clock-noise calibration to be formed entirely from hardware-generated data. The goal of this work is to realize a laboratory-scale analog of the LISA inter-

satellite metrology system—a “miniLISA”—that bridges the gap between idealized software simulations and the flight instrument, and provides a controlled environment for testing the robustness of TDI to instrumental non-idealities and realistic gravitational-wave signal injection.

The goal is for both noise sources, once processed through TDI and clock-noise calibration, to be reduced below the level required for gravitational-wave signal recovery across the core LISA measurement band.

The article is divided as follows:

II. LISA INTERFEROMETRIC DETECTION SYSTEM

We now describe the LISA interferometric detection system (IDS) and present a mathematical model of the noises. We specifically focus on relevant elements for laser frequency noise and USO jitter propagation and coupling.

A. Interferometric Phase Measurements

Each of the three identical LISA satellites carries two lasers on board. We refer to one as the local laser and the other as the adjacent laser. These two laser fields interfere on the test-mass and reference interferometers on the two optical benches of the spacecraft. In addition, the local laser interferes with the laser received from a distant spacecraft on the science interferometer. It is this science interferometer that encodes the gravitational-wave signal by measuring fluctuations in the spacecraft-to-spacecraft distance, while the test mass interferometer will measure the spacecraft-to-test mass displacement. In Figure 1 (a), a simplified LISA two-way armlink involving only the science interferometer is shown.

We begin by writing the optical phase of the laser field originating from movable optical system assembly (MOSA) ij on spacecraft i as a function of a common modelling time t :

$$\Phi_{ij}(t) = \nu_{ij}t + \delta\phi_{ij}(t), \quad (1)$$

where ν_{ij} denotes the nominal carrier frequency and $\delta\phi_{ij}(t)$ represents the intrinsic laser phase noise. Throughout this section, phases are expressed in cycles, frequencies in Hz, and clock errors in seconds. Capital Φ denotes total phase, while $\delta\phi$ denotes phase fluctuations about the nominal linear evolution. Nominal carrier and modulation frequencies are assumed constant. Laser phase fluctuations carry two MOSA indices here, $\delta\phi_{ij}$. The local clock reads $t_i = t + q_i(t)$, where q_i is its timing error and $\dot{q}_i$ its fractional-frequency error. We retain linear noise couplings with constant nominal beat frequencies. A relativistic model would instead distinguish each spacecraft’s proper time $\tau_i(t)$ from the common time t ,

writing $t_i = \tau_i(t) + q_i(\tau_i(t))$. This introduces time transformations in the propagation delays and frequency conversions, in addition to instrumental clock noise [4]; these effects are outside the present testbed model.

Each of the laser fields is phase-modulated using an electro-optic modulator (EOM) driven by the local USO. This is used to imprint sidebands on the laser field that carry the USO timing noise $q_i(t)$, with modulation phase noise $\nu_{ij}^m q_i(t)$. Since there is only one USO per spacecraft, clock noise carries a single spacecraft index.

The modulation frequency ν_{ij}^m will be about 2.4 GHz, which is upconverted from the USO via a phase locked loop (PLL) synthesizer, preserving its noise with high fidelity. For the adjacent and opposing MOSAs ik and ji respectively, the corresponding modulation frequency differs by 1 MHz, so 2.401 GHz, such that the sideband–sideband beat notes are offset by 1 MHz from the carrier–carrier beat note. The modulation depth $m = 0.53$ balances the optical power allocated to the carrier and the first-order sidebands. Figure 2 shows how the USO signal is distributed to the modulation, sampling-clock, and pilot-tone synthesis chains. Their role in clock-noise suppression is discussed in subsection II C. Lower frequency offset sidebands will also be used for pseudo-random ranging and data transfer between the satellites, but are not covered here.

As the beam propagates from MOSA ij toward spacecraft j and its MOSA ji , it accumulates phase contributions from Doppler shifts due to the relative spacecraft motion and from gravitational waves integrated along the arm. Following the separation of background propagation and GW perturbations in [4], we write

$$d_{ji}^{\text{tot}}(t) = d_{ji}(t) + H_{ji}(t). \quad (2)$$

Here $d_{ji}(t)$ is the background propagation time, which can vary slowly with the constellation breathing, and $H_{ji}(t)$ is its GW-induced perturbation. The first index denotes the receiving spacecraft and the second the emitting spacecraft. In the common-time model used here, this is a propagation-time decomposition; we do not include the transformations between spacecraft proper times contained in the proper pseudorange of [4]. We define the delay operator using only the background propagation time,

$$\mathbf{D}_{ji}x(t) = x(t - d_{ji}(t)). \quad (3)$$

Applying the total propagation time to Eq. (1) and expanding to first order gives

$$\begin{aligned} \Phi_{ij}(t - d_{ji}(t) - H_{ji}(t)) &\simeq \nu_{ij} [t - d_{ji}(t)] \\ &\quad + \mathbf{D}_{ji}\delta\phi_{ij}(t) - \nu_{ij}H_{ji}(t), \end{aligned}$$

where the product $H_{ji}\mathbf{D}_{ji}\delta\phi_{ij}$ of the GW perturbation and laser-frequency noise has been neglected. We therefore introduce the GW phase contribution

$$\delta\phi_{ji}^{\text{GW}}(t) = -\nu_{ij}H_{ji}(t). \quad (4)$$

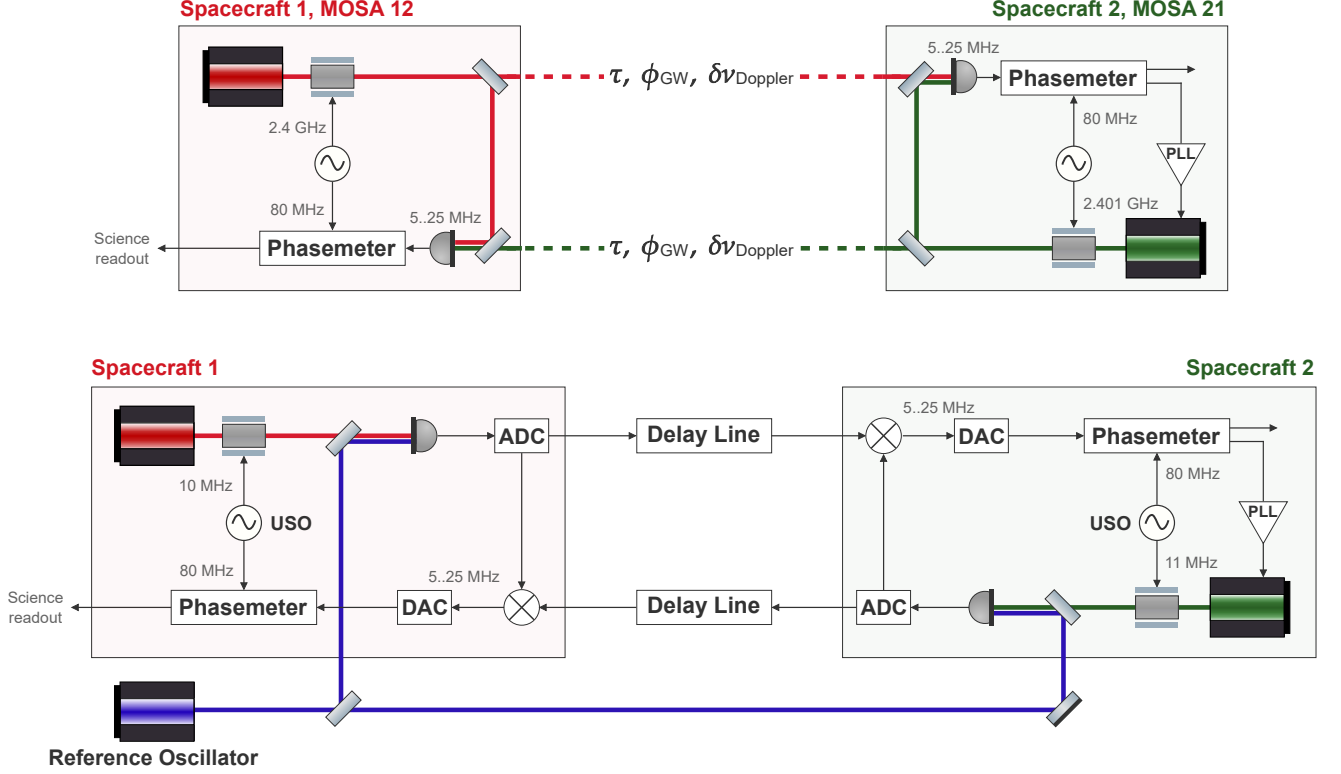


FIG. 1. A comparison of the two-way links as implemented in LISA (upper) and the testbed (lower). The upper panel shows the conceptual LISA configuration: the laser on Spacecraft 1 (MOSA 12) is phase-modulated at approximately 2.4 GHz using the local USO and transmitted to Spacecraft 2 (MOSA 21). The received beam interferes with the local laser, producing heterodyne beat notes in the MHz range that contain gravitational-wave phase fluctuations, Doppler shifts due to orbital motion, and laser frequency noise. These beat notes are tracked by the phasemeter, which is clocked by the onboard 80 MHz sampling reference derived from the USO. The lower panel depicts the laboratory implementation of the same signal topology. Instead of physical light propagation over millions of kilometres, the inter-spacecraft delay is emulated electronically using digital delay lines between ADC and DAC stages. A common reference oscillator is used to downconvert the laser fields from hundreds of terahertz to tens of megahertz, which can then be sampled by the delay line ADCs. The actual LISA-like heterodyne beat notes are created in the mixers.

To this order, the received total phase is $\mathbf{D}_{ji}\Phi_{ij}(t) + \delta\phi_{ji}^{\text{GW}}(t)$: the GW signal is an additive phase contribution, separate from the background delay operator. For constant ν_{ij} , the corresponding frequency fluctuation is $\delta\dot{\phi}_{ji}^{\text{GW}} = -\nu_{ij}\dot{H}_{ji}$.

At spacecraft j , the received field i interferes with the local laser field of phase $\Phi_{ji}(t)$ on a photodetector. The resulting heterodyne signal contains both the carrier-carrier beat note and the sideband-sideband beat notes. Additional cross terms lie outside the heterodyne range of the phasemeter, 5 to 25 MHz.

The electrical signal from the photodetector is then digitized by the phase measurement system (PMS) front-end electronics. The sampling process introduces a coupling of the local USO timing error $q_j(t)$ into the measured phase. For all measurement readouts, we adopt the LISA convention that phases are written as “distant

minus local.”¹ We define the nominal carrier coefficient

$$\alpha_{ji} = \Delta\nu_{ji} = \nu_{ij} - \nu_{ji}. \quad (5)$$

Since a sample labelled t_j is taken at $t \simeq t_j - q_j(t_j)$, sampling adds the phase error $-\alpha_{ji}q_j$. Keeping the nominal beat phase, the sampled carrier phase is, with the time argument again denoted by t ,

$$\begin{aligned} \Phi_{ji}^c(t) = & \alpha_{ji}t - \nu_{ij}d_{ji}(t) \\ & + \mathbf{D}_{ji}\delta\phi_{ij}(t) - \delta\phi_{ji}(t) \\ & + \delta\phi_{ji}^{\text{GW}}(t) - \alpha_{ji}q_j(t). \end{aligned} \quad (6)$$

Here the nominal phase $\alpha_{ji}t - \nu_{ij}d_{ji}(t)$ includes the background propagation contribution. Its derivative is

¹ The phasemeter measures the absolute beat frequency; its sign is assigned using the frequency plan.

$\alpha_{ji} - \nu_{ij}\dot{d}_{ji}(t)$; using α_{ji} alone in the sampling term neglects the Doppler contribution to the clock-coupling coefficient. The signal is then passed to digital phase lock loops (DPLLs), which track the carrier and both sideband beat notes. In the following readouts, the nominal phase evolution is removed from the tracked phase, leaving the fluctuations used for GW detection and clock-noise suppression. To complete the carrier readout model, we now include optical-bench motion and additive noise from the optical paths and readout electronics. The carrier phase measurement recorded on spacecraft j for light received from spacecraft i is

$$\begin{aligned} \text{sci}_{ji}^c(t) = & \mathbf{D}_{ji}\delta\phi_{ij}(t) - \delta\phi_{ji}(t) - \alpha_{ji}q_j(t) \\ & + \delta\phi_{ji}^{\text{GW}}(t) + N_{ji}^c(t) \\ & - \frac{\nu_0}{c}(\mathbf{n}_{ij} \cdot \mathbf{D}_{ji}\mathbf{\Delta}_{ij}(t) + \mathbf{n}_{ji} \cdot \mathbf{\Delta}_{ji}(t)), \end{aligned} \quad (7)$$

The term $N_{ji}^c(t)$ represents combined readout noise from optical path-length fluctuations and electronics. The vectors $\mathbf{\Delta}_{ij}(t)$ are optical-bench displacements in metres. The unit vector $\mathbf{n}_{ji}$ points along the incoming beam from spacecraft i to spacecraft j , and $\mathbf{n}_{ij} = -\mathbf{n}_{ji}$ in the geometry used for these displacement terms. The factor $\nu_0/c = 1/\lambda_0$ converts path-length fluctuations to phase in cycles, with $\nu_0 \simeq 281$ THz for the nominal wavelength $\lambda_0 = 1064$ nm. MHz-scale offsets between lasers are small compared with this optical frequency, so a common ν_0 suffices for the displacement terms. Those offsets remain essential in the heterodyne clock coefficients, which depend on frequency differences.

The corresponding upper sideband-sideband beat note readout can be written as

$$\begin{aligned} \text{sci}_{ji}^{\text{sb},+}(t) = & \mathbf{D}_{ji}\delta\phi_{ij}(t) - \delta\phi_{ji}(t) - \gamma_{ji}q_j(t) \\ & + \delta\phi_{ji}^{\text{GW}}(t) + N_{ji}^{\text{sb},+}(t) \\ & - \frac{\nu_0}{c}(\mathbf{n}_{ij} \cdot \mathbf{D}_{ji}\mathbf{\Delta}_{ij}(t) + \mathbf{n}_{ji} \cdot \mathbf{\Delta}_{ji}(t)) \\ & + \nu_{ij}^m \mathbf{D}_{ji}q_i(t) - \nu_{ji}^m q_j(t). \end{aligned} \quad (8)$$

The sideband clock coupling coefficient is

$$\gamma_{ji} = \Delta\nu_{ji} + \nu_{ij}^m - \nu_{ji}^m = \Delta\nu_{ji} + \Delta\nu_{ji}^m = \Delta\nu_{ji}^{\text{sb},+}. \quad (9)$$

Again, the term $N_{ji}^{\text{sb},+}(t)$ represents stochastic readout noise specific to the upper sideband measurement. The lower sideband has the same laser phase noise but the opposite sign for each modulation contribution. The sidebands have optical frequencies $\nu_{ij} \pm \nu_{ij}^m$, so the same path-length change produces a slightly different phase shift in each sideband. Using ν_0/c for their displacement couplings neglects a fractional correction of order $\nu^m/\nu_0 \simeq 9 \times 10^{-6}$. Similarly, the upper-sideband GW phase is $-(\nu_{ij} + \nu_{ij}^m)H_{ji}$, differing from the carrier by $-\nu_{ij}^m H_{ji}$. We neglect this small difference in the readout equations above and use the same $\delta\phi_{ji}^{\text{GW}}$ for the carrier

and both sidebands. The δ denotes a perturbation to the nominal optical phase, rather than stochastic noise.

For completeness, we also write out the reference and test mass interferometer readings on MOSA ji

$$\begin{aligned} \text{tm}_{ji}(t) = & \delta\phi_{jk}(t) - \delta\phi_{ji}(t) + \beta_{ji}q_j(t) + \mu_j(t) + N_{ji}^{\text{tm}}(t) \\ & + \frac{2\nu_0}{c}\mathbf{n}_{ji} \cdot (\mathbf{\delta}_{ji}(t) - \mathbf{\Delta}_{ji}(t)), \end{aligned} \quad (10)$$

$$\text{ref}_{ji}(t) = \delta\phi_{jk}(t) - \delta\phi_{ji}(t) + \beta_{ji}q_j(t) + \mu_j(t) + N_{ji}^{\text{ref}}(t). \quad (11)$$

Here we have introduced the fiber noise μ_j coming from the optical fiber that exchanges the local and adjacent laser fields between MOSA ji and MOSA jk . This term is typically referred to as the backlink noise. We have assumed it to be independent of the direction of propagation of the optical beams within. The vector $\mathbf{\delta}_{ji}$ is the test-mass displacement in metres, including displacement produced by acceleration noise; it is not itself an acceleration. With the same sampling convention, the reference and test-mass clock coefficient is

$$\beta_{ji} = \nu_{ji} - \nu_{jk}. \quad (12)$$

It has the opposite sign to the adjacent-minus-local beat frequency, which explains the plus sign multiplying $\beta_{ji}q_j$ in the readouts.

The full LISA noise budget also includes tilt-to-length coupling, which is not represented in our testbed model. The testbed omits the test-mass and reference interferometers and does not reproduce optical-bench motion, backlink fiber noise, or test-mass acceleration noise; these terms are therefore omitted in the following sections.

B. Laser Noise and TDI

In the LISA measurement system, laser frequency noise is first reduced onboard by locking one laser to an optical resonator and phase-locking the remaining lasers to this pre-stabilized reference with fixed frequency offsets. The last step ensures that all heterodyne beat notes remain within the phasemeter bandwidth while maintaining coherent phase relationships between spacecraft. However, residual laser frequency noise remains many orders of magnitude above the displacement sensitivity requirement and must be suppressed in post-processing. TDI achieves this by forming linear combinations of one-way phase measurements and appropriately delayed copies thereof, effectively synthesizing an equal-arm interferometer in post-processing.

For scale, consider a representative pre-stabilized frequency-noise amplitude of $30 \text{ Hz}/\sqrt{\text{Hz}}$ [4]. At $f = 10 \text{ mHz}$, the corresponding single-laser phase-noise amplitude is $30/(2\pi f) \simeq 480 \text{ cycles}/\sqrt{\text{Hz}}$, equivalent to about $0.5 \text{ mm}/\sqrt{\text{Hz}}$ at 1064 nm. This illustrative conversion, before any cancellation between measurements,

is many orders of magnitude above picometre-scale displacement noise.

First-generation TDI combinations cancel laser noise for constant light-travel times. Orbital motion makes these delays time-dependent, so their order of application matters and the cancellation is no longer exact. Second-generation combinations use longer sequences of delays to cancel the leading residual from arm-length changes, leaving terms involving higher powers of the arm velocities and arm accelerations [5].

Accurate knowledge of the delays is essential, as an error δd in the applied delay leads to residual laser noise proportional to the time derivative of the laser phase noise. This sets the requirement on delay knowledge to approximately one meter, motivating LISA's dedicated ranging system.

Before TDI is implemented, intermediary variables are constructed. The reference and test-mass interferometer readings remove optical-bench displacement noise, as discussed in [5, 6]. To make the signs explicit, define

$$T_{ji} = \text{tm}_{ji} - \text{ref}_{ji},$$

$$\xi_{ji} = \text{sci}_{ji}^c - \frac{1}{2}(T_{ji} + \mathbf{D}_{ji}T_{ij}).$$

Within the common-optical-frequency approximation above, the bench terms cancel, leaving the test-mass contribution

$$M_{ji} = -\frac{\nu_0}{c} [\mathbf{n}_{ji} \cdot \boldsymbol{\delta}_{ji}(t) + \mathbf{D}_{ji}(\mathbf{n}_{ij} \cdot \boldsymbol{\delta}_{ij}(t))].$$

The following equations omit readout noises. Reciprocal backlink noise cancels in the local reference difference

$$z_{ji} = \frac{1}{2}(\text{ref}_{ji} - \text{ref}_{jk}) = \delta\phi_{jk} - \delta\phi_{ji} + \beta_{ji}q_j.$$

For cyclic triples $(i, j, k) = (1, 2, 3), (2, 3, 1), (3, 1, 2)$, the intermediate variables can then be defined as

$$\begin{aligned} \eta_{ji} &= \xi_{ji} - z_{ji} \\ &= \mathbf{D}_{ji}\delta\phi_{ij}(t) - \delta\phi_{jk}(t) - (\alpha_{ji} + \beta_{ji})q_j(t) \\ &\quad + \delta\phi_{ji}^{\text{GW}}(t) + M_{ji}(t), \\ \eta_{ij} &= \xi_{ij} + \mathbf{D}_{ij}z_{ji} \\ &= \mathbf{D}_{ij}\delta\phi_{jk}(t) - \delta\phi_{ij}(t) - \alpha_{ij}q_i(t) \\ &\quad + \mathbf{D}_{ij}[\beta_{ji}q_j(t)] \\ &\quad + \delta\phi_{ij}^{\text{GW}}(t) + M_{ij}(t). \end{aligned}$$

Here M_{ij} is obtained by exchanging i and j in M_{ji} . These combinations reduce the number of independent laser phase noises from six to three, retaining $\delta\phi_{12}$, $\delta\phi_{23}$ and $\delta\phi_{31}$, while removing the optical-bench displacement terms. The construction introduces an asymmetry between the two link orientations.

Further, as an example, a commonly used first-generation Michelson-type TDI observable synthesised at spacecraft i is the X_i combination, which can be written

as

$$\begin{aligned} X_i(t) &= \left[(\eta_{ik}(t) + \mathcal{D}_{ik}\eta_{ki}(t)) + \mathcal{D}_{ik}\mathcal{D}_{ki}(\eta_{ij}(t) + \mathcal{D}_{ij}\eta_{ji}(t)) \right] \\ &\quad - \left[(\eta_{ij}(t) + \mathcal{D}_{ij}\eta_{ji}(t)) + \mathcal{D}_{ij}\mathcal{D}_{ji}(\eta_{ik}(t) + \mathcal{D}_{ik}\eta_{ki}(t)) \right], \end{aligned} \quad (13)$$

where $\mathcal{D}_{ij}$ is the delay operator – the cursive indicates it is being applied in post-processing.

C. Clock-noise Coupling and Correction

The second-largest noise contribution in the raw measurements is timing jitter arising from the USOs [red book]. Similarly to laser frequency noise, it exceeds the sensitivity requirement by several orders of magnitude and must be removed in post-processing before gravitational wave signals can be recovered. Unlike laser frequency noise, it couples into the measurement only locally at one spacecraft. Hence, to still allow for a similar differential measurement of the clock noise, the laser fields exchanged between the satellites are modulated with a signal derived from the USOs. From the upper and lower sideband phase measurements at the science interferometer, as written in (8), a correcting term can be constructed.

Using the phase-domain counterpart of the clock-transfer construction in [7], and neglecting readout and modulation noises initially, we define

$$r_{ji} = \frac{\text{sci}_{ji}^{\text{sb},+}(t) - \text{sci}_{ji}^c(t)}{\nu_{ij}^m} = \mathbf{D}_{ji}q_i(t) - q_j(t). \quad (14)$$

The retained GW phase cancels in this difference because we have approximated the carrier and sideband GW couplings as equal. Keeping their optical-frequency difference would instead add $-H_{ji}$ to r_{ji} ; the cancellation is therefore approximate. The normalization uses the transmitted modulation frequency ν_{ij}^m , giving r_{ji} units of seconds. Equivalently, $r_{ji} = (\text{sci}_{ji}^{\text{sb},+} - \text{sci}_{ji}^{\text{sb},-}) / (2\nu_{ij}^m)$ under the same approximations. These variables contain delayed remote clock noise minus local clock noise, matching the structure of laser noise in the science readouts. Delayed combinations of the r variables can therefore be used to construct clock corrections for the TDI observables [7]. For clock noise removal to work properly, the modulation must track the effective measurement-clock reference. Let $N_{ij}(t)$ denote differential timing noise in the tone derivation for MOSA ij , in seconds, and let $N_{ij}^m(t)$ denote additional optical modulation phase noise, in cycles. The excess upper-sideband phase is

$$m_{ij}(t) = \nu_{ij}^m N_{ij}(t) + N_{ij}^m(t).$$

Including these terms and readout noise gives

$$r_{ji} = \mathbf{D}_{ji}q_i - q_j + \frac{\mathbf{D}_{ji}m_{ij} - m_{ji}}{\nu_{ij}^m} + \frac{N_{ji}^{\text{sb},+} - N_{ji}^c}{\nu_{ij}^m}.$$

Although both N_{ij}^m and the readout terms $N_{ji}^{\text{sb},+}$ are phase noises, they enter at different points. Modulation noise is imprinted on the emitted optical field and follows it through the local and delayed paths, whereas readout noise is added when each beat note is measured. Keeping them separate preserves their different delays and correlations in the clock correction. The local tone-derivation noise also carries the factor $-\nu_{ji}^m/\nu_{ij}^m$, which cannot be replaced by unity when the modulation frequencies differ. These excess noises propagate through the clock correction into the corrected measurements. The distinction between timing and direct phase noise also matters for the modulation-frequency choice. Referred to the source modulation frequency, $m_{ij}/\nu_{ij}^m = N_{ij} + N_{ij}^m/\nu_{ij}^m$. Increasing the transmitted and local modulation frequencies by the same factor therefore suppresses a fixed direct phase-noise contribution to r_{ji} , but not a fixed differential timing noise; the local timing term retains the ratio ν_{ji}^m/ν_{ij}^m . A measurement of the combined modulation noise at a single frequency does not by itself separate these contributions.

The ultimate clock reference will be the pilot tone, which is also derived from the USO and is used to correct ADC sampling jitter within the PMS. The differential timing jitter between the modulating frequency of 2.4 GHz and the corresponding pilot tone will be below 50 fs/ $\sqrt{\text{Hz}}$ as per requirements. For the adjacent 2.401 GHz, this requirement is relaxed as the digital synthesis from the 10 MHz USO can not be as noise-free. This additional noise is called 'modulation noise' in [7]. Hence, the corresponding clock noise correcting r -variables themselves need correcting. For this purpose, LISA reference interferometers also measure an onboard sideband-sideband beat note. From these, similar correcting variables as (14) are defined and can be used to remove the accessible differential modulation-noise contribution.

III. MINILISA DESIGN

A. Motivation for the Present Work

The present work focuses on an optical extension of the FPGA-based delay-line architecture, combining physical laser noise, independently running spacecraft (SC) clocks with sideband-based clock transfer, and second-scale delays within a single measurement framework. The electrical delay-line implementation and its TDI demonstrations are described separately [REF: electrical delay-line work]. Here, we describe the optical input and the full

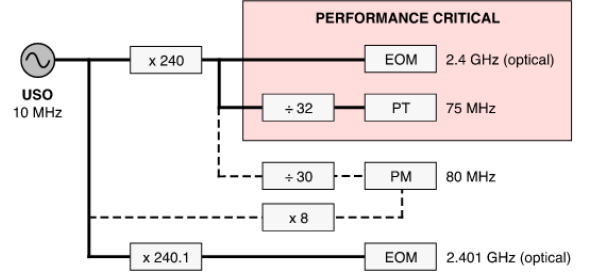


FIG. 2. Clock-generation and distribution scheme highlighting performance-critical paths. A 10 MHz USO signal is multiplied to 2.4 GHz for optical sideband modulation via the EOM, forming the primary clock-transfer tone. Parallel synthesis chains generate the 75 MHz PT used for ADC timing correction and the 80 MHz sampling clock driving the PM. The shaded region marks the performance-critical elements, where clock phase noise must be preserved with high fidelity. A slightly offset 2.401 GHz modulation is generated for the opposite spacecraft to produce MHz-offset sideband-sideband beat notes. The architecture ensures coherent distribution of clock noise into both optical modulation and digital sampling paths, enabling subsequent clock-noise suppression in post-processing.

testbed measurement model, and use component characterization to assess the expected performance of the combined setup. The two-arm configuration represents three SC with one laser per SC; a third inter-spacecraft arm is a possible extension.

The second-scale delays are arguably the most consequential missing element. In LISA, the inter-spacecraft light-travel times enter the TDI combinations as applied delays, and any error in the delay knowledge leaves a residual laser noise proportional to the laser noise derivative. Accurate delay estimation is therefore not a secondary concern but a fundamental requirement for TDI performance. More broadly, the interaction between realistic delays and the digital processing chain can produce effects that, while analytically predictable, are only fully quantified in hardware. The non-commutativity of the delay operator and the phasemeter CIC decimation filters is one such example: predicted analytically in 2019 [REF] and subsequently confirmed and characterized experimentally by the LOT [REF]. In a testbed with hardware-implemented second-scale delays, such couplings are present by construction and can be studied under realistic signal conditions.

B. Measurement Philosophy

The objective of the testbed is not to reproduce every physical subsystem of LISA, but rather to reproduce the mathematical structure of the primary noises and the GW signal. From the perspective of the data analysis chain, the essential requirement is that laser frequency

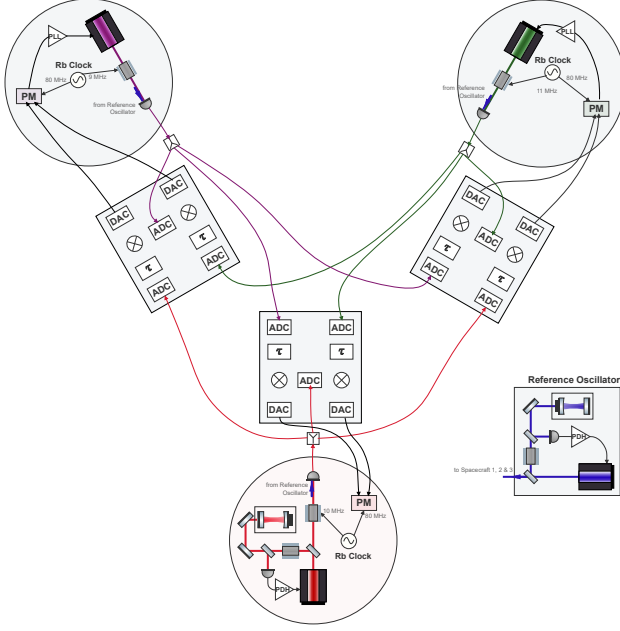


FIG. 3. Triangular LISA testbed laboratory implementation. Three simulated spacecraft are arranged in a closed topology, each equipped with a laser, phasemeter (PM), and an independent Rb clock. The MHz heterodyne beat notes are digitised and routed through programmable delay blocks that emulate the inter-spacecraft light-travel times required for TDI. In general, two signals are delayed on each delay-line board; these are configured to represent the incoming links to a given spacecraft, while the clock driving that delay board is matched to the corresponding spacecraft phasemeter. A common optical reference oscillator provides pre-stabilisation via PDH locking. This architecture preserves the essential structure of LISA’s one-way measurements while replacing physical arm lengths with electronic delays.

noise and clock noise enter the measurements in the same form as they do in LISA. The first index of a measurement or delay denotes the receiving SC. Since the testbed uses one laser and one modulator per SC, source quantities carry a single SC index: $\delta\phi_i$, q_i , and ν_i^m .

$$\eta_{ij}(t) = \mathbf{D}_{ij}\delta\phi_j(t) - \delta\phi_i(t) + \delta\phi_{ij}^{\text{GW}}(t),$$

while clock noise enters as a local coupling proportional to the heterodyne frequency,

$$\eta_{ij}(t) = \mathbf{D}_{ij}\delta\phi_j(t) - \delta\phi_i(t) + \delta\phi_{ij}^{\text{GW}}(t) - \alpha_{ij}q_i(t).$$

Additionally, the sideband measurements contain delayed clock noise from the remote SC,

$$\eta_{ij}^{\text{sb},+}(t) = \mathbf{D}_{ij}\delta\phi_j(t) - \delta\phi_i(t) + \delta\phi_{ij}^{\text{GW}}(t) - (\alpha_{ij} + \nu_j^m)q_i(t) + \nu_j^m\mathbf{D}_{ij}q_j(t),$$

which is used for building the clock-noise correcting variables. Here α_{ij} is the signed nominal carrier beat frequency; the upper-sideband coefficient is $\gamma_{ij} = \alpha_{ij} +$

$\nu_j^m - \nu_i^m$. The equations above show the ideal signal and noise-coupling structure, using the same GW phase for the carrier and sidebands under the approximation stated in the IDS section. The additional modulation, board, and readout noises are introduced below. We therefore define a measurement channel as *LISA-like* if it reproduces these coupling structures.

In addition to getting the noises right, incorporating LISA-like delays on the order of seconds is what makes the TDI validation physically meaningful: without realistic light-travel times, the dominant error sources — ranging inaccuracy, delay-operator non-commutativity, and clock synchronization initialization — can not be properly exercised. For this, we use an electronic delay line implemented on an FPGA. This means we need to convert the original laser fields from the 281 THz frequency to a range measurable by the electronics. We do this by heterodyning each laser field with a reference laser. This way, the phase information in the mHz range is preserved, and the reference laser phase noise enters as part of the effective laser noise removed by TDI, as shown below. The FPGA-based delay line is also capable of injecting the phase response to gravitational waves and Doppler shifts.

The measurement is built up of links like the one shown in the lower diagram of Figure 1. Each satellite is represented by one laser and an independent clock. We neglect the test mass and reference interferometers. We do not consider test mass acceleration noise, optical bench jitters, backlink noise, or tilt-to-length coupling.

C. Optical Input

To incorporate realistic laser frequency noise, the proposed setup uses three lasers — each SC represented by one laser. The three lasers are hereafter referred to as the SC lasers. The optical setup also supports frequency-offset locking between lasers, analogous to the offset locking foreseen in LISA, and a clock tone modulation scheme. Compared to the real mission, this simplified configuration assumes perfect phase locking between the two lasers on each SC and neglects backlink noise and differential optical bench jitter. This reproduces the laser-noise coupling of LISA’s three-noise η observables after the local reference measurements are combined, with the additional noise from those measurements and the backlink neglected here. A fourth laser — the reference laser — serves as a common heterodyne source for all three. All four lasers are Ten64 units from VM Photonics, chosen for their low frequency and relative intensity noise. The role of the reference laser is to convert the optical phase fluctuations of each SC laser into electronically processable heterodyne beat notes, which can then be measured by the delay line. This heterodyning step effectively downconverts the optical signals from approximately 281 THz to tens of megahertz, preserving laser frequency noise in the millihertz range of interest. Among

the SC lasers, one is frequency-stabilized to a 25 cm ultra-low expansion (ULE) glass cavity, while the reference laser is locked to a 30 cm ULE cavity via the Pound-Drever-Hall method. Both cavities are housed in a vacuum chamber surrounded by three layers of aluminum thermal shielding, which suppresses thermal fluctuations sufficiently to achieve LISA-like pre-stabilized laser frequency noise (LFN) levels of around $30 \text{ Hz}/\sqrt{\text{Hz}}$ in the millihertz band. The remaining two SC lasers are PLL-locked to the synthetic LISA-like beat notes. Each SC laser is phase-modulated using a fiber-coupled broadband EOM driven at a frequency derived from the same independent USO that clocks the corresponding phasemeter ADC. To enable a wider range of modulation frequencies, the reference laser is modulated as well, producing sideband-sideband beat notes on the photodetector. Otherwise, the electronic delay line ADCs pose a limitation on the modulation frequency. This modulation is currently used only for clock tone transfer. The beat notes entering the delay lines are therefore not themselves LISA-like; the final LISA-like observables are synthesised downstream using digital mixers. The construction of a two-way arm link is illustrated in Fig. 1(b), alongside a simplified schematic of the corresponding LISA link showing only the functionally relevant elements. Taking the signed photodetector beat phase as SC laser minus reference laser, the carrier phase can be written as

$$\Phi_{\text{PD},i}^{\text{c}}(t) = \nu_{\text{R},i}t + \delta\phi_i(t) - \delta\phi_{\text{R}}(t),$$

where $\nu_{\text{R},i}$ is the signed nominal beat note frequency between the reference laser and the SC i laser. The upper-sideband phase is

$$\begin{aligned} \Phi_{\text{PD},i}^{\text{sb},+}(t) = & (\nu_{\text{R},i} + \nu_i^m)t \\ & + \delta\phi_i(t) - \delta\phi_{\text{R}}(t) + \nu_i^m q_i(t) \\ & + \nu_i^m N_i(t) + N_i^m(t). \end{aligned}$$

Here, N_i is the differential timing noise between the modulation and the effective SC measurement-clock reference, while N_i^m is the additional modulation phase noise. These are the single-index counterparts of N_{ij} and N_{ij}^m in the IDS section. We retain both contributions and write their combined upper-sideband phase noise as

$$m_i(t) = \nu_i^m N_i(t) + N_i^m(t).$$

This noise is added to the modulation before the signal is split between the boards. It therefore follows the SC sideband through the local and delayed paths. The lower sideband carries the opposite sign of the modulation phase, including $\nu_i^m q_i$ and m_i .

Our modulation-noise measurement constrains the combined phase noise m_i at the measurement frequency; it does not distinguish N_i from N_i^m . The distinction matters when predicting the effect of increasing the modulation frequency. With respect to timing, the excess is

$$\frac{m_i}{\nu_i^m} = N_i + \frac{N_i^m}{\nu_i^m}.$$

A fixed direct phase-noise contribution is reduced by a higher modulation frequency in this timing representation, whereas a fixed timing-jitter contribution is not. This scaling assumes that the underlying noise itself remains unchanged when the modulation frequency is changed.

Photodetection and its electronics introduce additive noise from relative intensity fluctuations, shot noise and detector electronics. This contributes to the phase estimated by the delay-line PLLs. For the performance estimate, we treat the resulting readout contributions as uncorrelated between measurement paths, without taking credit for possible cancellation of shared noise by TDI. Their equivalent phase noise is included together with the uncorrelated electronic noise in the final readout terms below.

The photodetector output is then split and distributed across three delay-line boards. One of these boards synthesizes the local LISA-like beat note signals, the other two do so for the other two SC. The weaving of the photodetector signal between the delay line boards is illustrated in Figure 3.

D. The Delay Line

A central component of the testbed is the digital delay line, which emulates the light-travel time of the LISA arms. It can apply time shifts roughly up to 8.3 s to a signal and its two sidebands—the duration corresponding to the 2.5 million km inter-spacecraft distance in LISA. This value is currently limited by the amount of RAM memory. The precise delay, controllable at the nanosecond level, can be continuously updated during runtime based on simulated LISA orbital data, for example, computed using the *lisaorbits* Python package. In addition to the delay, the system can impose the corresponding Doppler frequency shifts on both the carrier and sidebands, and more importantly, it can inject gravitational wave (GW) signals—either single templates or multiple waveforms simultaneously—into the data stream.

The delay line is implemented entirely on an AMD Xilinx ZCU208 evaluation board, which integrates a Zynq™ UltraScale+™ RFSoc FPGA with high-speed 14-bit ADCs/digital to analog converter (DAC)s and an embedded processing unit (PS) capable of running a Linux-based operating environment. This architecture allows the FPGA's programmable logic to apply precise time delays and Doppler shifts in real time, while the PS computes and updates the required parameters and handles clock-domain synchronization between the delay logic and the external timing system.

As a part of the testbed signal chain, the delay line boards accept an electronic input signal — a MHz-frequency beat note — which is digitized by the onboard ADC. The digitized data stream is then passed to a PLL, which extracts the frequency information. This is moved to the RAM for buffering. Subsequently, Doppler shifts

and GW modulations are added digitally, producing delayed and frequency-shifted replicas of the original signal. These operations are performed in parallel for the carrier and sideband components. The background delay is applied to the tracked input frequency, and the GW contribution is subsequently added as a frequency offset before phase accumulation. The carrier and sideband components receive the same GW frequency increment. The mapping to $\delta\phi_{ij}^{\text{GW}}$ is described in the following subsection; the nominal Doppler evolution is omitted from the phase-fluctuation equations below. The delayed signal then gets electronically mixed with the local signal. Writing the effective input laser noise phase as

$$\delta\phi_i^{\text{eff}} = \delta\phi_i - \delta\phi_{\text{R}},$$

the ideal carrier output contains $\mathbf{D}_{ij}\delta\phi_j^{\text{eff}} - \delta\phi_i^{\text{eff}} + \delta\phi_{ij}^{\text{GW}}$. The board also introduces timing errors and additional phase noise. We keep the board timing error ϵ_i , in seconds, separate from the uncorrelated phase noise. In the simplified common-ADC/DAC-clock model, its carrier and upper-sideband contributions are

$$\begin{aligned}\delta\phi_{\epsilon,ij}^{\text{c}} &\simeq \nu_{\text{R},j}(1 - \mathbf{D}_{ij})\epsilon_i, \\ \delta\phi_{\epsilon,ij}^{\text{sb},+} &\simeq (\nu_{\text{R},j} + \nu_j^m)(1 - \mathbf{D}_{ij})\epsilon_i.\end{aligned}$$

Here $\nu_{\text{R},j}$ is the nominal electrical frequency of the remote input carrier. These expressions assume the same nominal input and reconstructed frequency for each component. The remaining uncorrelated board phase noise is combined with the optical and final phasemeter readout noise in b_{ij}^a below.

The measured delay-line baseline provides the electronic contribution to this combined noise budget. Contributions already included in that measurement are not added again separately, including any timing noise represented explicitly by ϵ_i . The delay-line architecture and electrical characterization are described in [REF: electrical delay-line work]; here, they enter as inputs to the optical testbed performance model.

E. GW Signal Injection

The GW phase defined in Eq. (4) is implemented electronically after the background delay. With one laser per SC, the required phase on the link from j to i is

$$\delta\phi_{ij}^{\text{GW}}(t) = -\nu_j H_{ij}(t).$$

Since the nominal optical frequency ν_j is constant, the frequency offset supplied to the phase accumulator is

$$\delta\nu_{ij}^{\text{inj}}(t) = \delta\dot{\phi}_{ij}^{\text{GW}}(t) = -\nu_j \dot{H}_{ij}(t). \quad (15)$$

Accumulating this offset reproduces the required GW phase up to an initial phase constant. The scale factor is the represented optical frequency, not the MHz beat note frequency entering the delay-line ADC. Thus

the GW signal is generated by adding phase evolution to the reconstructed signal; the hardware delay itself remains d_{ij} and is not additionally perturbed by H_{ij} . This implements the first-order propagation model of the IDS section without requiring the delay register to resolve the GW-induced change in light-travel time.

The waveform supplied for each directed link is evaluated at its reception time in the common modelling time coordinate. The propagation and source geometry are already included in H_{ij} , so no additional background delay is applied to the injection waveform after it is generated. Relative timing between the playback streams must reproduce this convention.

The same frequency offset is added to the carrier and both sideband phase accumulators in the present implementation. This realizes the common-GW-phase approximation used in the IDS equations. Resolving the relative sideband correction would instead require offsets proportional to $\nu_j \pm \nu_j^m$, but that correction is neglected here. Consequently, the prescribed GW phase cancels from the sideband-minus-carrier clock variable, while remaining in the individual science readouts and their TDI combinations.

The link responses can be computed in Python for a chosen waveform and constellation geometry. Multiple sources can be combined by summing their link responses in the linear regime. The source class is not fixed by the hardware, but the resulting commands must lie within its frequency range, resolution and update bandwidth. A set of independently prescribed link signals can also be used for calibration; a GW realization requires the relative amplitudes and phases appropriate to a common source and geometry.

In the current delay-line implementation, a signed 32-bit GW word K_{ij} is added on the 64-bit phase-increment scale, giving

$$\delta\nu_{ij}^{\text{inj}} = K_{ij} \frac{f_{\text{clk}}}{2^{64}}.$$

For the nominal 125 MHz accumulator clock, this corresponds to approximately 6.78×10^{-12} Hz per least significant bit and an offset range of ± 0.01455 Hz. The frequency command is held between timestamped updates. Its realized phase accuracy therefore depends on command quantization, update timing and output phase reconstruction, rather than the offset-word width alone. These are injection-path properties to be characterized or taken from the electrical implementation's characterization [REF: electrical delay-line work]; the equations here describe the intended signal response.

F. Readout and Data Processing

The electrical signals produced by the delay line are passed to a phasemeter for digitization and phase extraction. In the current implementation, we use a Moku:Pro phasemeter. In future iterations, this will be replaced by

a ZCU104 evaluation board with custom front-end electronics, allowing closer replication of LISA phasemeter features such as pilot-tone correction and digital filtering.

Now, at the level of the measurement model, the testbed corresponds to a LISA-like constellation in which there is no backlink noise and the two lasers on each SC are assumed to be perfectly phase-locked with zero offset. We further assume ideal reference and test-mass measurement and neglect optical-bench jitter. The optical modulation, board and readout noises introduced above are retained in the following measurements.

Under these assumptions, the science measurement replicated by the testbed, as illustrated in Fig. 1, can be written as

$$\eta_{ij}(t) = \mathbf{D}_{ij}\delta\phi_j^{\text{eff}}(t) - \delta\phi_i^{\text{eff}}(t) + \delta\phi_{ij}^{\text{GW}}(t) - \alpha_{ij}q_i(t) + \nu_{\text{R},j}(1 - \mathbf{D}_{ij})\epsilon_i(t) + b_{ij}^c(t),$$

and the corresponding upper-sideband-upper-sideband beat note as

$$\begin{aligned} \eta_{ij}^{\text{sb},+}(t) &= \mathbf{D}_{ij}\delta\phi_j^{\text{eff}}(t) - \delta\phi_i^{\text{eff}}(t) + \delta\phi_{ij}^{\text{GW}}(t) - \gamma_{ij}q_i(t) \\ &\quad + \nu_j^m \mathbf{D}_{ij}q_j(t) - \nu_i^m q_i(t) \\ &\quad + \mathbf{D}_{ij}[\nu_j^m N_j(t) + N_j^m(t)] \\ &\quad - \nu_i^m N_i(t) - N_i^m(t) \\ &\quad + (\nu_{\text{R},j} + \nu_j^m)(1 - \mathbf{D}_{ij})\epsilon_i(t) + b_{ij}^{\text{sb},+}(t). \end{aligned}$$

Here the final phasemeter contributes the sampling-clock terms $-\alpha_{ij}q_i$ and $-\gamma_{ij}q_i$, while the board timing error enters explicitly through ϵ_i . The terms b_{ij}^a , with $a \in \{\text{c}, \text{sb}, +\}$, collect the equivalent optical readout noise, the uncorrelated board phase noise and the additional phase-readout noise of the final phasemeter. All are expressed in cycles at the final link readout; the explicit timing and modulation terms are excluded. For the performance estimate, these additive phase noises are modelled as uncorrelated between distinct link and carrier/sideband readouts. The same b_{ij}^a is retained wherever that readout is reused in TDI and clock correction. This combined term plays the role of the additive readout noise in the IDS section. The effective SC timing error q_i and the board timing error ϵ_i may be correlated through their reference distribution and are not assumed to be independent.

The reference laser phase noise enters through

$$\mathbf{D}_{ij}\delta\phi_j^{\text{eff}} - \delta\phi_i^{\text{eff}} = \mathbf{D}_{ij}\delta\phi_j - \delta\phi_i - (\mathbf{D}_{ij} - 1)\delta\phi_{\text{R}}.$$

It therefore has the same delayed-difference structure as the effective SC laser noises and is treated by TDI in the same manner. The reference laser also introduces correlations between those effective noises. A quieter reference reduces its contribution, but independence of the effective laser noises is not required for the algebraic cancellation. Optical readout noises and phase-locking correlations are accounted for separately in the noise budget.

Finally, the upper-sideband-minus-carrier variable takes the form

$$\begin{aligned} r_{ij} &= \frac{\eta_{ij}^{\text{sb},+} - \eta_{ij}}{\nu_j^m} \\ &= \mathbf{D}_{ij}q_j - q_i + \frac{\mathbf{D}_{ij}m_j - m_i}{\nu_j^m} \\ &\quad + (1 - \mathbf{D}_{ij})\epsilon_i + \frac{b_{ij}^{\text{sb},+} - b_{ij}^c}{\nu_j^m}. \end{aligned}$$

The additive GW phase cancels in this difference because it is common to carrier and sideband. The additional terms show how modulation noise, board timing jitter and the combined additive phase noise enter the clock correction as timing errors. In particular,

$$\begin{aligned} \frac{\mathbf{D}_{ij}m_j - m_i}{\nu_j^m} &= \mathbf{D}_{ij}N_j - \frac{\nu_i^m}{\nu_j^m}N_i \\ &\quad + \frac{\mathbf{D}_{ij}N_j^m - N_i^m}{\nu_j^m}. \end{aligned}$$

Increasing both modulation frequencies at fixed ratio suppresses the last line for fixed direct phase noises, but does not suppress the timing-jitter terms in the first line. Since the present modulation measurement does not separate these contributions, both possibilities must be retained when projecting the performance of a higher-frequency implementation. Their propagation through TDI and clock correction is considered in the performance section.

IV. PERFORMANCE MODELLING

We estimate the instrumental noise contributions by combining component measurements with a calculation of the optical readout noise and propagating these inputs through the measurement equations. The component comparison and the predicted readout spectra are shown in Figs. 4 and 5. These estimates describe the secondary noises available to the clock-corrected observable under ideal laser- and clock-noise cancellation. They do not constitute a measurement of the sensitivity of the assembled optical testbed.

A. Component Noise and Comparison Reference

We express phase fluctuations in cycles and use one-sided power spectral densities (PSDs), denoted by S . The plotted amplitude spectral densities (ASDs) are $\sqrt{S}$, in cycles/ $\sqrt{\text{Hz}}$. The measured modulation spectrum is a phase PSD and is therefore converted to ASD by taking its square root, without an additional frequency-dependent conversion.

The modulation-noise estimator is formed from the upper- and lower-sideband phase measurements,

$$\hat{m}(t) = \frac{\phi_U(t) - \phi_L(t)}{2}. \quad (16)$$

The opposite modulation signs in the two sidebands allow the common optical phase to cancel in this difference. The resulting spectrum constrains the combined modulation phase noise $m_i = \nu_i^m N_i + N_i^m$ at the measurement operating point, together with differential readout contributions to the estimator. It does not independently determine the timing and direct phase-noise terms. We use the first phase-PSD column of `measured noises/modulator_psd.csv`; the other PSD columns are diagnostic sideband differences. In the illustrative propagation calculation, the measured spectrum is assigned to three independent spacecraft modulation-noise sources.

The electronic input is the measured delay-line residual from `measured noises/baseline.csv`, shown in Fig. 4. This spectrum is obtained after forming a delayed phase difference and applying the measurement's timing correction. It is thus a characterization of that measurement chain, rather than a direct determination of every individual board noise. Assigning its PSD to each final carrier and sideband readout is a provisional model of the electronic contribution. Its interpretation must retain the operating conditions and the noise contributions already present in the measurement. Both measured inputs support the plotted 0.25 mHz–1 Hz band without extrapolation.

For comparison, we use the single-link reference adopted in the component analysis. With $\lambda = 1064$ nm, its equivalent phase ASDs are

$$A_{\text{OMS}}(f) = \frac{15 \times 10^{-12} \text{ m}}{\lambda} \sqrt{1 + \left(\frac{2 \text{ mHz}}{f}\right)^4}, \quad (17)$$

$$A_{\text{acc}}(f) = \frac{3 \times 10^{-15} \text{ m s}^{-2}}{(2\pi f)^2 \lambda} \sqrt{1 + \left(\frac{0.4 \text{ mHz}}{f}\right)^2} \times \sqrt{1 + \left(\frac{f}{8 \text{ mHz}}\right)^4}. \quad (18)$$

The reference is $A_{\text{link}} = (A_{\text{OMS}}^2 + A_{\text{acc}}^2)^{1/2}$. Acceleration noise is included here as a comparison quantity; it is not an experimentally present test-mass noise in the current testbed. The component spectra are displayed directly, without applying a TDI transfer function. Consequently, this figure compares input noise scales and does not establish a gravitational-wave sensitivity.

B. Predicted Carrier and Sideband Readout Noise

For the readout calculation, we consider one phase-modulated laser interfering with an unmodulated reference laser on the Newport photodetector. Each beam delivers 8 mW at the detector, after any combining losses,

and the phase-modulation depth is $\beta = 0.53$ rad. This operating mode provides an explicit carrier-carrier and sideband-carrier readout estimate. A configuration with both lasers modulated, as discussed in the optical design, requires the corresponding sideband-sideband beat amplitudes to be substituted; the present numerical values should not be transferred to that mode without this adjustment.

For optical powers P_{SC} and P_{ref} , responsivity $\mathcal{R}$, and power-overlap efficiency η_{het} , the peak current of beat order n is

$$I_{n,\text{pk}} = 2\mathcal{R}\sqrt{\eta_{\text{het}}P_{\text{SC}}P_{\text{ref}}}|J_n(\beta)|. \quad (19)$$

The carrier uses $n = 0$, and either first-order sideband uses $|n| = 1$. Here $J_0(0.53) = 0.9310$ and $J_1(0.53) = 0.2558$. The modulated beam therefore contains 6.93 mW in its carrier and 0.523 mW in each first-order sideband. Using $\mathcal{R} = 0.6$ A/W and $\eta_{\text{het}} = 0.9$ gives peak beat currents of 8.48 mA and 2.33 mA, respectively.

We include shot noise, dark-current shot noise, Johnson noise, relative intensity noise (RIN), and ADC quantization noise. Their one-sided current ASDs are modelled as

$$i_{\text{shot}} = \sqrt{2e[\mathcal{R}(P_{\text{SC}} + P_{\text{ref}}) + I_d]}, \quad (20)$$

$$i_J = \sqrt{4k_B T / R_L}, \quad (21)$$

$$i_{\text{RIN}} = \mathcal{R}(P_{\text{SC}} + P_{\text{ref}})A_{\text{RIN}}, \quad (22)$$

$$i_{\text{ADC}} = \frac{V_{\text{FS}}}{2^{B_{\text{ADC}}} G \sqrt{6} f_s}. \quad (23)$$

The adopted parameters are $I_d = 100$ nA, $T = 300$ K, $R_L = 50 \Omega$, $G = 50$ V/A, $V_{\text{FS}} = 1$ V, $B_{\text{ADC}} = 14$ and $f_s = 2$ GHz. The RIN ASD is $A_{\text{RIN}} = 0.9 \times 10^{-8} / \sqrt{\text{Hz}}$. The RIN expression retains a conservative coherent sum of the two beams' intensity fluctuations. It represents additive noise around the RF beat frequencies; low-frequency amplitude modulation is not assumed to couple directly to phase. The ADC term assumes white quantization noise, AC coupling and no clipping, rather than a measured effective number of bits.

For additive noise that is locally white around the selected beat, demodulation combines noise from both RF sidebands. The equivalent phase ASD of contribution k is

$$A_{\phi,n}^{(k)} = \frac{\sqrt{2} i_k}{2\pi I_{n,\text{pk}}}. \quad (24)$$

The factor 2π converts radians to cycles. No integration over the PLL bandwidth is applied to these spectral densities. Summing the component PSDs gives carrier and first-sideband readout ASDs of 2.78×10^{-9} and 1.01×10^{-8} cycles/ $\sqrt{\text{Hz}}$, respectively, equivalent to 17.5 and 63.7 nrad/ $\sqrt{\text{Hz}}$. Their ratio is $J_0(\beta)/|J_1(\beta)| \simeq 3.64$. These are predictions for the stated optical powers and white-noise assumptions, not measurements of the complete readout chain.

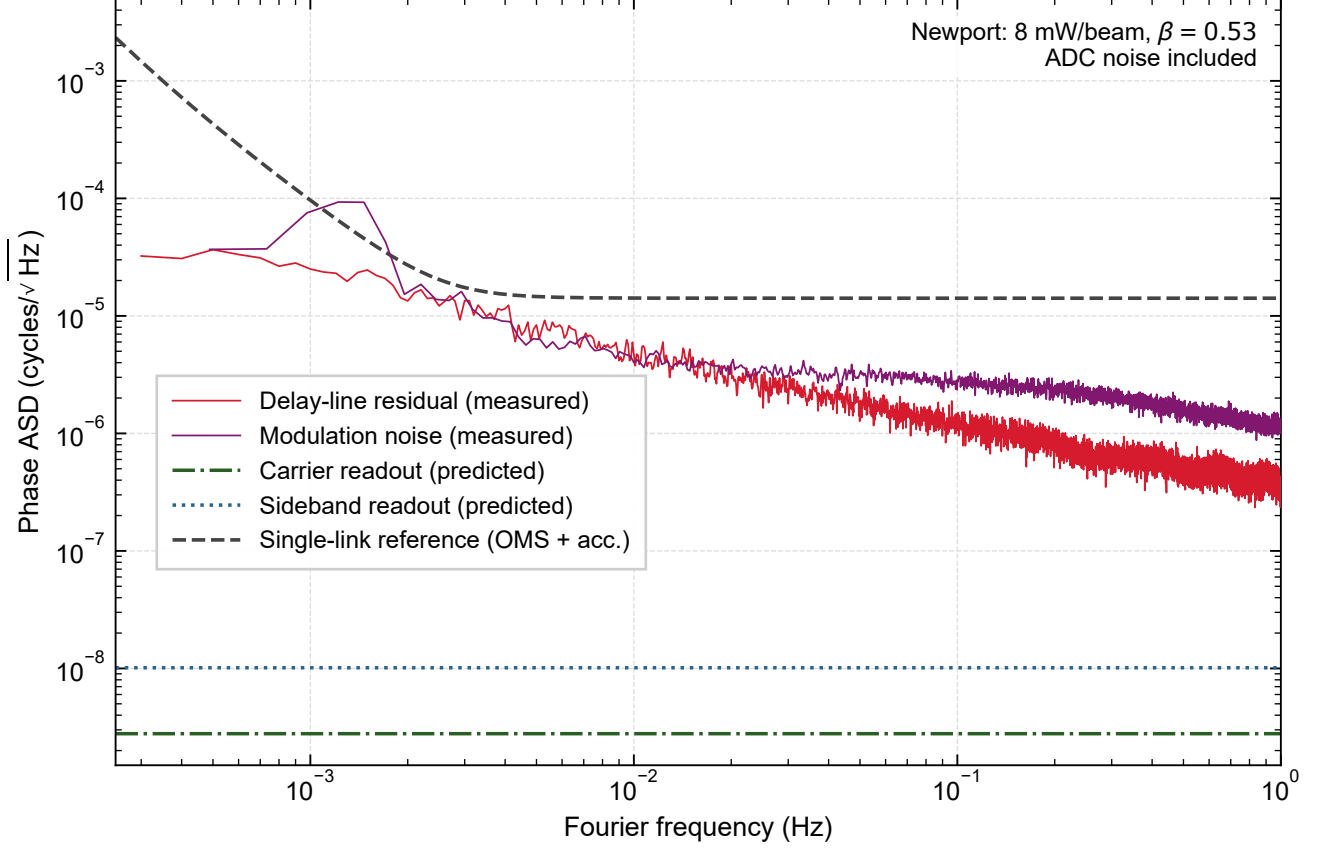


FIG. 4. Component phase-noise amplitude spectral densities compared with the single-link reference formed from optical metrology and acceleration noise. Solid curves show the measured delay-line residual and modulation phase-noise estimator; measured Fourier bins are plotted without smoothing. The predicted Newport carrier and first-order sideband readout noises include shot noise, dark-current shot noise, Johnson noise, relative intensity noise and ADC quantization noise. The optical model assumes one phase-modulated beam and one unmodulated reference beam, each delivering 8 mW at the detector, with modulation depth 0.53 rad. The detector responsivity is 0.6 A/W, heterodyne efficiency is 0.9, and the termination is 50 Ω at 300 K. The ADC model uses a 1 V full-scale span, 14 bits and 2 GS/s. The assumed white RIN is $0.9 \times 10^{-8}/\sqrt{\text{Hz}}$, summed coherently between the beams. The model uses 35 MHz modulation and electrical carrier frequencies of 15.0, 10.9 and 21.6 MHz; the readout prediction assumes white noise across the corresponding RF bands. All curves are shown before propagation through TDI and clock correction; the measured electronic residual is shown as characterized and is not deconvolved into individual board noise sources. The comparison is not a measurement of the complete testbed sensitivity.

C. Propagation Through TDI and Clock Correction

The propagation calculation uses static delays and constant nominal frequencies. The illustrative frequency plan is $\nu_{R,i} = (15.0, 10.9, 21.6)$ MHz and $\nu_i^m = 35$ MHz for all three spacecraft, giving $(\alpha_{12}, \alpha_{21}, \alpha_{13}, \alpha_{31}) = (-4.1, 4.1, 6.6, -6.6)$ MHz. The nominal one-way delay is $L = 2.5 \times 10^9 \text{ m}/c \simeq 8.339 \text{ s}$. These specify the model operating point; they are not a reconstruction of the frequencies used in every component measurement.

Writing $z_{ij} = \exp(-2\pi i f d_{ij})$, $A = z_{12}z_{21}$ and $B =$

$z_{13}z_{31}$, the static Michelson defined above becomes

$$X = (1 - A)(\eta_{13} + z_{13}\eta_{31}) - (1 - B)(\eta_{12} + z_{12}\eta_{21}). \quad (25)$$

Let P_{ij} denote its four link coefficients and K_{ij} the clock-correction coefficients, chosen so that

$$X_c = \sum_{ij} (P_{ij}\eta_{ij} + K_{ij}r_{ij}) \quad (26)$$

has zero coefficient for each q_i . The sum runs over links 12, 21, 13 and 31, using the upper-sideband-minus-carrier definition of r_{ij} given above. The static second-generation convention used in the calculation is $X_{2c} =$

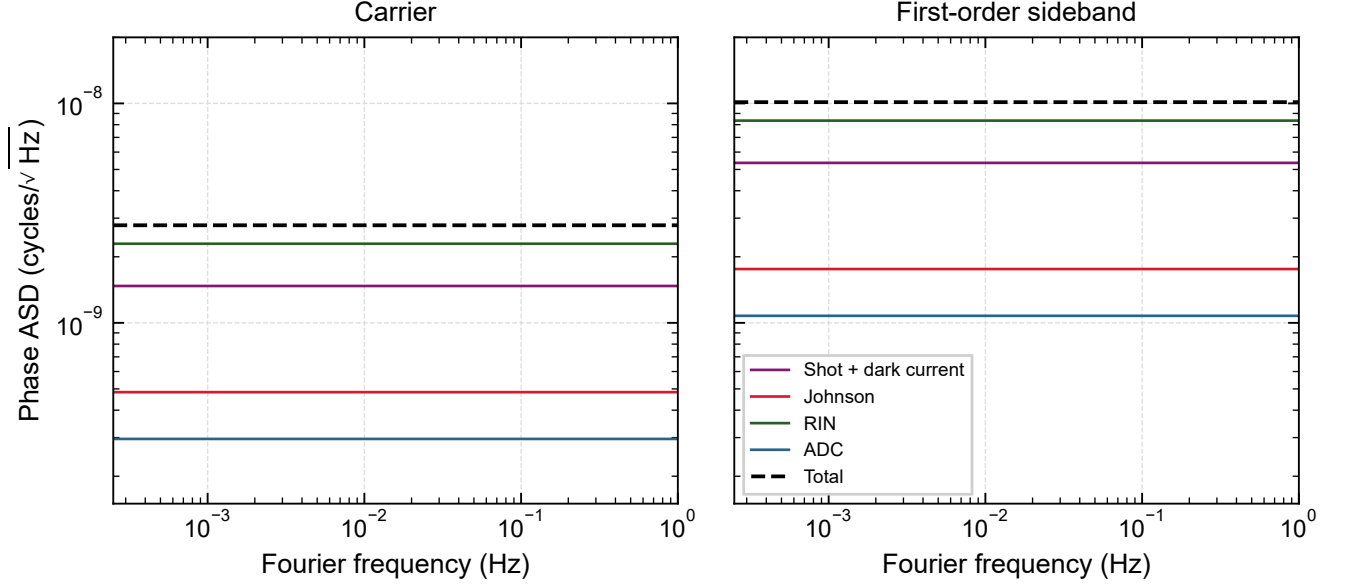


FIG. 5. Predicted Newport carrier and first-order sideband readout noise for one modulated and one unmodulated beam, each delivering 8 mW at the detector, with modulation depth 0.53 rad. The two panels share a common vertical scale. Shot noise includes dark-current shot noise; the ADC contribution is the white quantization estimate. The dashed curves are the quadrature totals. The different beat amplitudes in Eq. (19) give different carrier and sideband phase-noise levels even though their input current-noise spectra are taken to be equal.

$(1 - AB)X_c$; this extra filter must multiply both the science and correction weights. Laser and clock cancellation can be checked algebraically for unequal static delays. This static calculation does not model flexing arms or synchronization and interpolation errors.

Noise sources are retained coherently wherever a measurement is reused. In particular, for the first-generation weights the carrier and sideband readout coefficients are

$$T_{b_{ij}^c} = P_{ij} - \frac{K_{ij}}{\nu_j^m}, \quad T_{b_{ij}^{sb,+}} = \frac{K_{ij}}{\nu_j^m}. \quad (27)$$

The two appearances of a given carrier noise are therefore combined before taking a squared magnitude. For independent physical sources n_a , the output PSD is

$$S_{X_{2c}}(f) = \sum_a |T_a(f)|^2 S_a(f). \quad (28)$$

Where source correlations are retained, this sum is replaced by the corresponding spectral-covariance expression. The reference laser is one shared physical source, not an independent noise added to each effective laser.

Modulation noise survives clock correction. At fixed nominal beat frequencies, increasing all ν_i^m by a common factor s reduces its transfer coefficients by $1/s$. The output modulation ASD therefore decreases by $1/s$ for fixed direct phase noise. For fixed timing noise, the input modulation phase ASD increases by s , leaving the residual unchanged. This is distinct from laser-noise cancellation. Likewise, a small raw TDI ASD at low Fourier

frequency partly reflects the differencing response of the observable, which also acts on a gravitational-wave signal; it is not by itself an improvement in sensitivity.

D. Interpretation of the Estimate

Figure 6 shows the predicted total after static second-generation TDI and clock correction. The electronic baseline, modulation noise, and separate Newport carrier and sideband readout spectra are propagated through their respective corrected-observable transfer functions. Their output PSDs are added to form the total. The reference curve is the single-link noise model propagated as four independent equivalent link phase noises through the same TDI combination; it is not an unfiltered single-link curve. This estimate assumes ideal primary-noise cancellation and excludes an additional board-timing contribution and processing-error residuals for which no measured bounds have been supplied.

For the specified optical operating point, the calculated Newport readout spectra lie below the measured electronic and modulation component spectra over the plotted band. This indicates that the predicted detector readout is not the leading component limitation under these assumptions. Whether increasing the modulation frequency improves the corrected observable depends on the relative modulation and additive-readout contributions, and on whether the measured excess behaves as timing or direct phase noise. A GHz implementation also

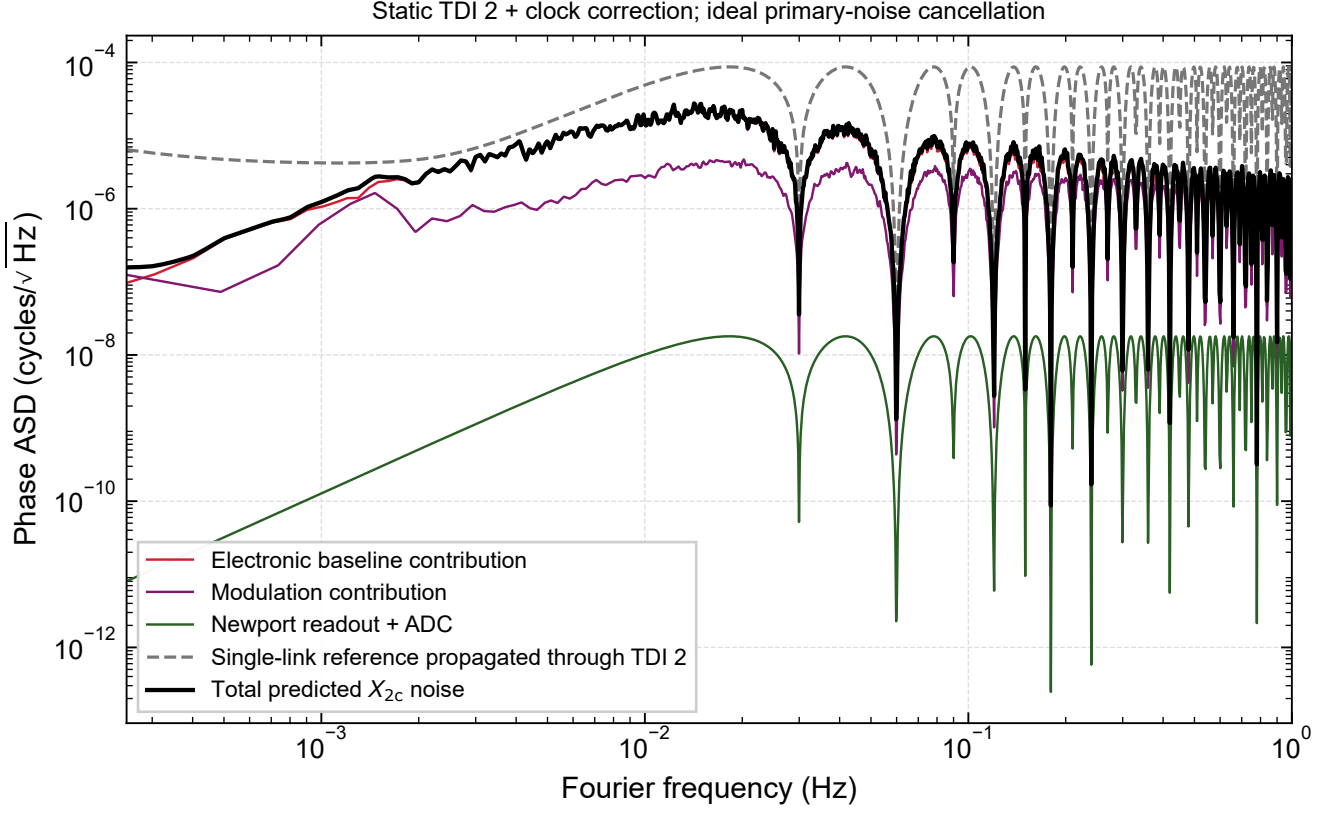


FIG. 6. Predicted phase-noise ASD of the clock-corrected static second-generation Michelson observable X_{2c} . The black curve is the quadrature total of the electronic-baseline, modulation and Newport readout contributions, including ADC quantization noise. Carrier readout noise is retained consistently in both the science measurement and the clock correction. The calculation uses 8.339 s one-way delays, 35 MHz modulation on all spacecraft, and the frequency plan specified in the text. The dashed curve is the single-link OMS-plus-acceleration reference propagated through the same TDI observable under an independent-equivalent-link assumption. The transfer-function nulls are properties of the observable, not broadband elimination of secondary noise. The total is conditional on treating the measured electronic residual and the predicted readout terms as separate contributions and on ideal laser- and clock-noise cancellation.

requires a compatible RF and digitization chain; holding the readout spectrum fixed while changing modulation frequency is only a model extrapolation.

A numerical total formed from the measured baseline and the predicted readout terms remains conditional on their separation. The explicit ADC estimate must not be counted again if the same ADC noise is already present in the baseline. Similarly, readout contamination of the modulation estimator must be accounted for before adding an independent estimate of that contribution. Board timing noise requires a measured spectrum or bound, and residual laser noise from imperfect delay knowledge requires a processing-error calculation. The figures therefore establish the component inputs and the predicted optical readout contribution; a validated total optical-testbed sensitivity requires these additional measurements and the consistent mapping of each measured residual into the final link equations.

V. FUTURE EXTENSIONS

A. Future Extensions: Injection of Acceleration Noise

B. Future Extensions: Three Satellite Constellation

A natural extension of the current setup is the inclusion of a third inter-spacecraft link. Once the two-arm configuration has been validated, adding a third arm would enable tests of a broader class of TDI combinations and provide closer correspondence to the full LISA geometry. As the present testbed is constrained to one laser per spacecraft, extending the setup to three arms requires careful frequency planning to ensure that all heterodyne beatnotes remain within the bandwidth of the phaseme-

ter.

For tests of first-generation TDI with static delays, keeping the three arising beatnote frequencies within the LISA phasemeter range is expected to be straightforward. When considering second-generation TDI with Doppler shifts, however, maintaining all beatnotes within the nominal 7–23 MHz range may become challenging. For experimental TDI validation, one could increase the phasemeter bandwidth. Alternatively, additional frequency shifters could be introduced to provide greater flexibility in controlling the beatnote frequencies.

C. Future Extensions: Changes to the Clock Noise Transfer System

Currently, clock-noise removal performance is ultimately limited by modulation noise arising from the generation of the clock sidebands. In LISA, this limitation is associated with the GHz frequency conversion processes and the subsequent modulation. In the current testbed, this effect is expected to be less pronounced, as the spacecraft laser fields are modulated at MHz frequencies, simplifying the sideband generation chain.

A future upgrade would replace the MHz modulation with GHz modulation to more closely reflect LISA. Implementing this change would require modifications to the signal chain, as the delay line ADCs impose an upper frequency limit of approximately 120 MHz. Hence, the reference oscillator would also need to be modulated to enable downconversion of the sidebands, similarly to the carrier signals.

Another possible change is the inclusion of at least one more modulator for one or all spacecraft and the corresponding reference measurement. For LISA, the modulation frequency conversion for the right side MOSA is known to be noisier compared to the left side, as the conversion multiplication implemented digitally does not correspond to a power of two. Hence, in the clock noise removal process, the correcting variables as defined by equation (14) have to be corrected themselves using the purely local sideband-sideband measurements. To include this effect, our testbed would need at least one spacecraft with explicit left and right side separation. This could be realized by introducing a second modulator driven by a deliberately noisier signal.

VI. CONCLUSION

We have presented the design and current status of miniLISA, a testbed extending prior FPGA-based TDI demonstrations to include physically representative optical laser noise and independently running SC clocks within a single hardware framework. Unlike previous testbeds, which have each reproduced a subset of these elements — physical lasers without independent clocks or realistic delays (UFLIS), independent clocks without pro-

grammable delays (Hexagon), or realistic delays without physical optical generation (LISA on Table) — miniLISA combines real lasers, independent Rb-disciplined clocks, second-scale programmable delays, and a three-SC topology in one setup.

The current noise budget shows a mixed picture. Readout noise — relative intensity noise and shot noise on the photodetectors — is analytically predicted to remain below the 1 pm displacement allocation across the core LISA band, suggesting it will not be the dominant limitation, though this remains a provisional conclusion pending full system integration. Delay-board clock jitter and the residual clock-noise coupling, by contrast, currently exceed the allocation by several orders of magnitude in both the analytical model and initial measurements. We have identified the dominant mechanism — board clock noise entering through the ADC/DAC chain, uncorrected by TDI or the standard clock-noise-removal algorithm — and outlined a correction scheme based on cross-board reference measurements, along with two independent mitigations for the underlying modulation-noise limitation: balanced detection, and an upgrade to GHz-frequency sideband modulation that would restore LISA-representative signal-to-readout-noise ratios in the clock-correcting variables.

Several open questions remain under active investigation, including the extent to which the current single-board measurements are limited by ranging error rather than by the mechanisms modeled here, and the characterization of second-order couplings not yet captured in the present noise model. Planned extensions include the addition of a third inter-spacecraft arm to enable genuine three-signal TDI combinations rather than a two-way approximation, and the GHz modulation upgrade described above. Together, these developments aim to bring miniLISA’s intrinsic noise floor below LISA-representative levels across all reproduced channels, at which point the transfer-function and suppression-ratio comparisons presented here can serve as a genuine, physically-generated test of the LISA measurement model rather than a characterization of the testbed’s own limitations.

-
- [1] Benjamin P Abbott, Richard Abbott, Thomas D Abbott, Matthew R Abernathy, Fausto Acernese, Kendall Ackley, Carl Adams, Thomas Adams, Paolo Addresso, Rana X Adhikari, et al. Observation of gravitational waves from a binary black hole merger. *Physical review letters*, 116(6):061102, 2016.
- [2] Monica Colpi, Karsten Danzmann, Martin Hewitson, Kelly Holley-Bockelmann, Philippe Jetzer, Gijs Nelemans, Antoine Petiteau, David Shoemaker, Carlos Sopuerta, Robin Stebbins, Nial Tanvir, Henry Ward, William Joseph Weber, Ira Thorpe, Anna Dauriskikh, Atul Deep, Ignacio Fernández Núñez, César García Marirrodiga, Martin Gehler, Jean-Philippe Halain, Oliver Jennrich, Uwe Lammers, Jonan Larrañaga, Maïke Lieser, Nora Lützgendorf, Waldemar Martens, Linda Mondin, Ana Piris Niño, Pau Amaro-Seoane, Manuel Arca Sedda, Pierre Auclair, Stanislav Babak, Quentin Baghi, Vishal Baibhav, Tessa Baker, Jean-Baptiste Bayle, Christopher Berry, Emanuele Berti, Guillaume Boileau, Matteo Bonetti, Richard Brito, Riccardo Buscicchio, Gianluca Calcagni, Pedro R. Capelo, Chiara Caprini, Andrea Caputo, Eleonora Castelli, Hsin-Yu Chen, Xian Chen, Alvin Chua, Gareth Davies, Andrea Derdzinski, Valerie Fiona Domcke, Daniela Doneva, Irina Dvorkin, Jose María Ezquiaga, Jonathan Gair, Zoltan Haiman, Ian Harry, Olaf Hartwig, Aurelien Hees, Anna Heffernan, Sascha Husa, David Izquierdo, Nikolaos Karnesis, Antoine Klein, Valeriya Korol, Natalia Korsakova, Thomas Kupfer, Danny Laghi, Astrid Lamberts, Shane Larson, Maude Le Jeune, Marek Lewicki, Tyson Littenberg, Eric Madge, Alberto Mangiagli, Sylvain Marsat, Ivan Martin Vilchez, Andrea Maselli, Josh Mathews, Maarten van de Meent, Martina Muratore, Germano Nardini, Paolo Pani, Marco Peloso, Mauro Pieroni, Adam Pound, Hippolyte Quelquejay-Leclerc, Angelo Ricciardone, Elena Maria Rossi, Andrea Sartirana, Etienne Savalle, Laura Sberna, Alberto Sesana, Deirdre Shoemaker, Jacob Slutsky, Thomas Sotiriou, Lorenzo Speri, Martin Staab, Danièle Steer, Nicola Tamanini, Gianmassimo Tasinato, Jesus Torrado, Alejandro Torres-Orjuela, Alexandre Toubiana, Michele Vallisneri, Alberto Vecchio, Marta Volonteri, Kent Yagi, and Lorenz Zwick. LISA definition study report. Version Number: 1.
- [3] Gregory Weaver, Jeffrey Garstecki, and Samuel Reynolds. The performance of ultra-stable oscillators for the gravity recovery and interior laboratory (grail). In *Proceedings of the 42nd Annual Precise Time and Time Interval Systems and Applications Meeting*, pages 369–380, 2010.
- [4] Jean-Baptiste Bayle and Olaf Hartwig. Unified model for the LISA measurements and instrument simulations. 107(8):083019.
- [5] Massimo Tinto and Olaf Hartwig. Time-delay interferometry and clock-noise calibration. *Physical Review D*, 98(4):042003, 2018.
- [6] Markus Otto, Gerhard Heinzel, and Karsten Danzmann. Tdi and clock noise removal for the split interferometry configuration of lisa. *Classical and Quantum Gravity*, 29(20):205003, 2012.
- [7] Olaf Hartwig and Jean-Baptiste Bayle. Clock-jitter reduction in LISA time-delay interferometry combinations. 103(12):123027. Publisher: American Physical Society.